

Project 1: Energy, Emissions and Development

DATA 201 - Spring 2026

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Motivation

Energy use is an indicator of technological, economic, social, development.

Combustion of fossil sources has led to increasing CO₂ emissions.

QUESTIONS:

- 1- Is CO₂ emission quantitatively correlated with the technological and economic development of nations over the past 30 years?**
- 2. Have different regions of the world developed independent correlations?**
- 3. Is the use of renewable energy evidenced by a reduction in CO₂ production?**
- 4. Is methane emission following similar trends to CO₂?**

Datasets

Our World in Data: CO2 and methane emissions, share of emissions from the various sources.

<https://github.com/owid/co2-data>

World Bank - Patent applications as measure of technological advancement.

<https://data.worldbank.org/indicator/IP.PAT.RES>

Bias: sparse information for the less developed or more restrictive countries. 1990-2022 has a reasonable overlap of emissions and patent data for many countries.

Merged working dataset after cleaning:

7,194 rows and 82 columns - Countries: 218 - Years: 1990 to 2022

Variables of interest

Variable	Definition	Type
country	Name of the country	Categorical
continent	Name of continent	Categorical
year	Calendar year of observation	Quantitative
iso_code	ISO 3-letter country code	Categorical
population	Total population	Quantitative
gdp	Gross domestic product (constant 2011 USD)	Quantitative
co2	Annual CO2 emissions (million tonnes)	Quantitative
methane	Annual methane emissions (MtCO2e)	Quantitative
primary_energy_consumption	Total primary energy (TWh)	Quantitative
patent_applications	Resident patent filings (WIPO)	Quantitative
co2_per_unit_energy	Renewable energy proxy (kg CO2/kWh)	Quantitative

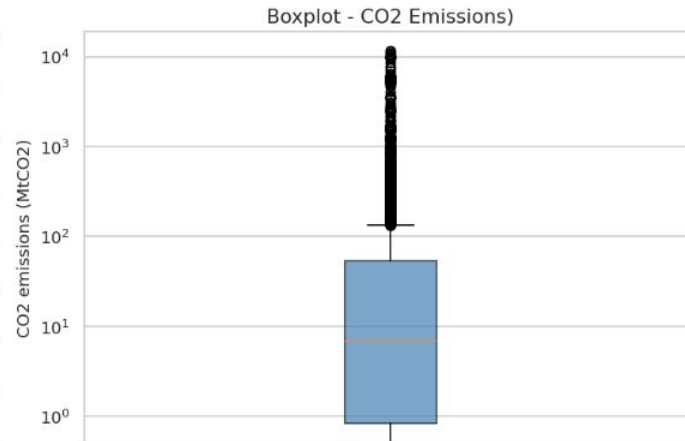
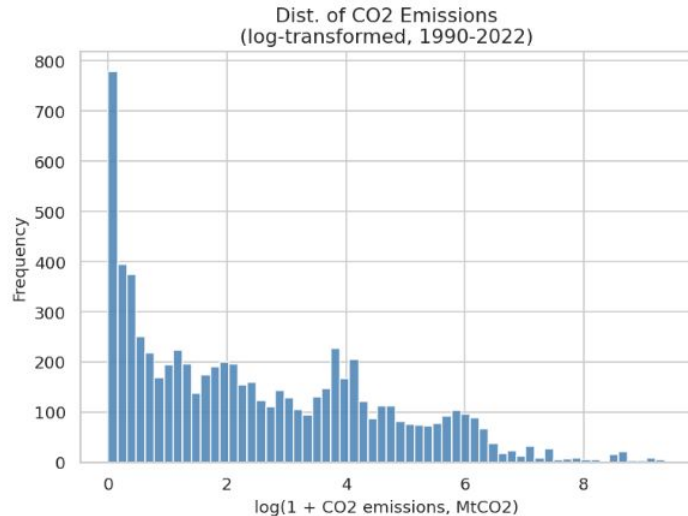
Analysis: Annual CO2 Emissions

Question: What is the distribution of annual CO2 emissions (in million tonnes) across country-years from 1990 to 2022, and what does that distribution reveal about global inequality in emissions?

Mean : 135.39 MtCO2

Median : 6.88 MtCO2

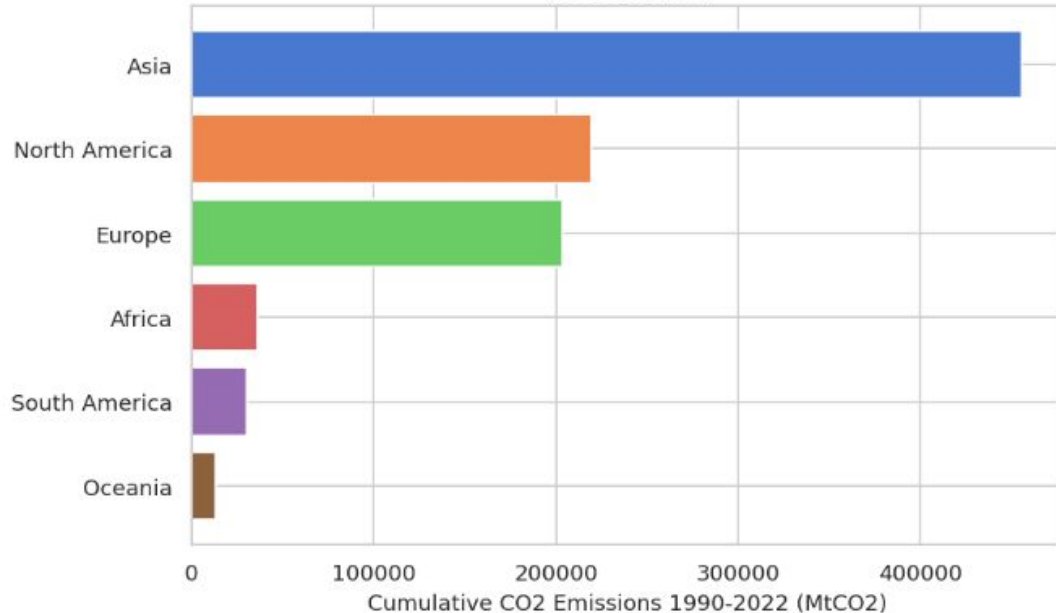
Std Dev: 656.52 MtCO2



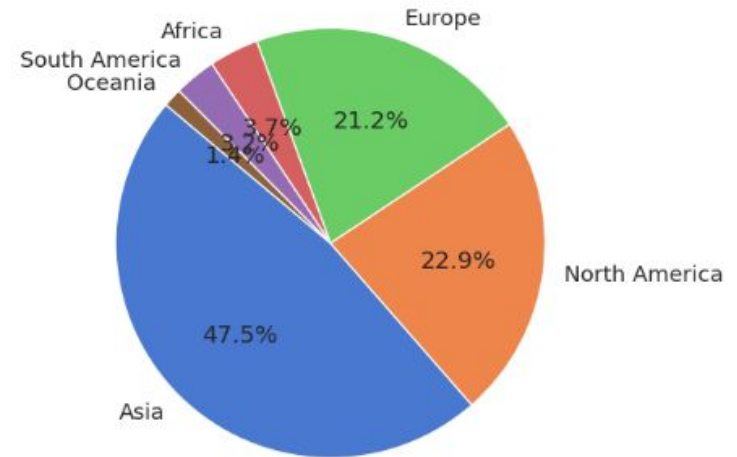
Analysis: Continent

Question: How are total CO2 emissions distributed across continents, and which ones contribute most to global emissions over 1990-2022?

Total CO2 Emissions by Region
(1990-2022)

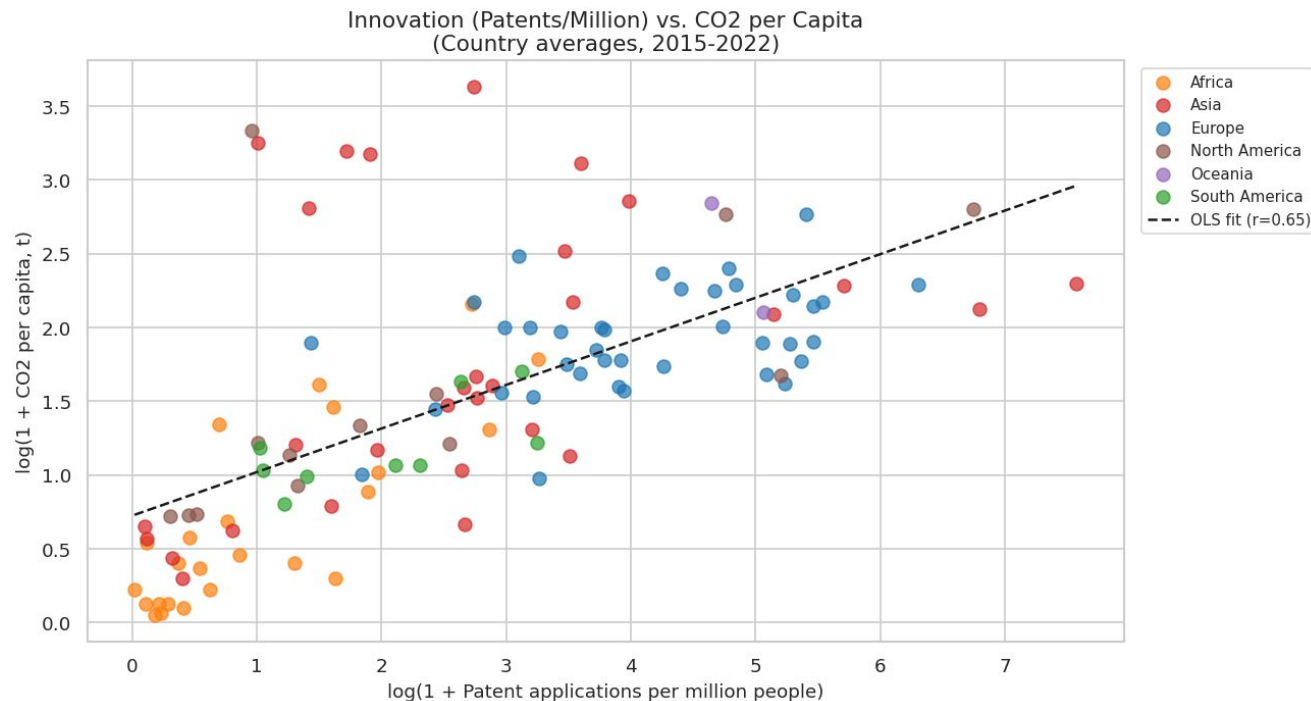


Share of Cumulative CO2 Emissions by Region



Relationship between Patents and CO2 Emissions (using 2015-2020 average)

Question: Is there a relationship between a country's technological innovation (resident patent filings per million people) and its annual CO2 emissions per capita? Does higher innovation correlate with lower or higher emissions intensity?*



	patent_per_million	co2_per_capita	gdp_per_capita
count	116.00	116.00	116.00
mean	78.78	5.56	22233.34
std	221.03	6.25	20656.51
min	0.02	0.05	685.77
25%	2.50	1.62	8129.50
50%	14.35	3.94	15232.27
75%	49.66	7.25	31519.44
max	1947.69	36.73	134206.56

Pearson r (log-log): 0.648

Nonparametric inference - bootstrap CI for median CO2/capita

Question: What is the median CO2 per capita among all country-years from 1990-2022, and how precisely can we estimate it from a 10% random sample?

Full population size $N = 7,016$

Population median = 2.7137 t CO2/person

Sample size $n = 702$ (10.0%)

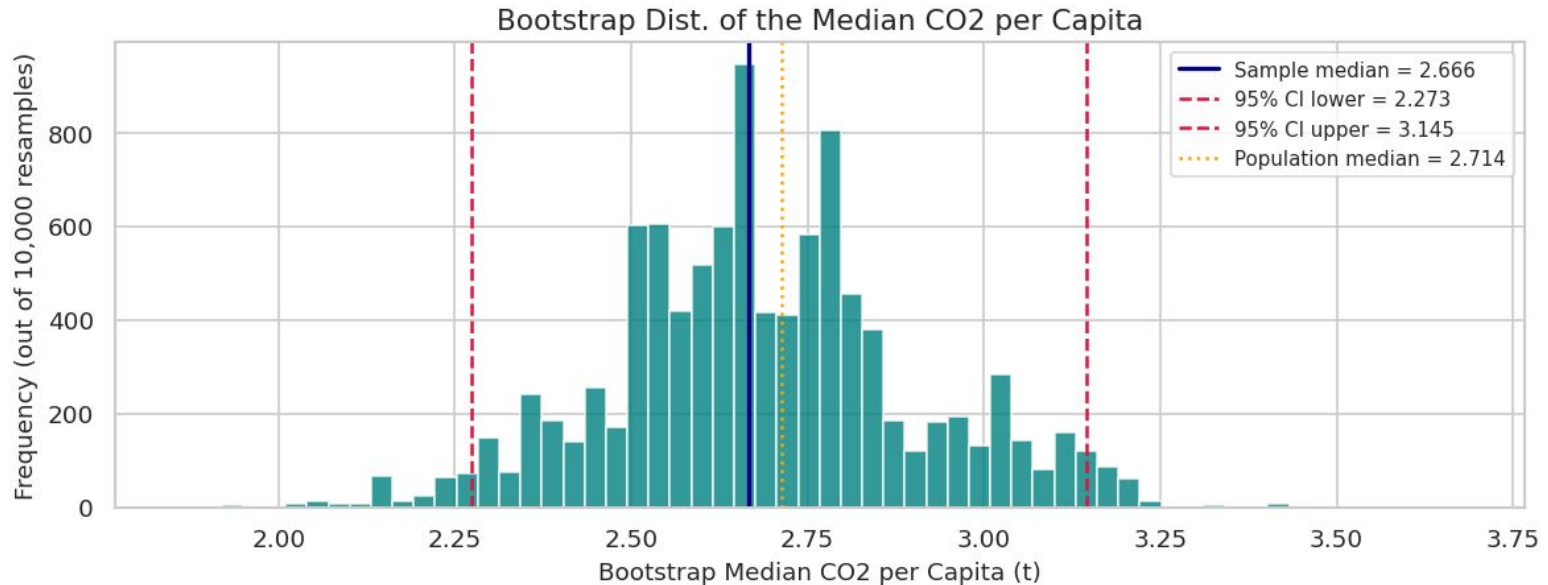
Sample median = 2.6660 t CO2/person

Bootstrap SE of median = 0.2160

95% CI (percentile method) =

[2.2734, 3.1448] t CO2/person

Interval width: 0.8714



Conclusions

Quantitative analysis (CO₂ emissions):

Annual CO₂ emissions are extremely right-skewed — a small number of large economies generate the vast majority of emissions.

Categorical analysis (Continent):

Asia accounts for the largest share of cumulative CO₂ emissions from 1990–2022, followed by Europe and North America. Africa contributes the least, despite being home to over 1.4 billion people — illustrating profound inequity in the historical emissions record.

Exploratory relationship (Patents vs. CO₂ per capita):

Higher patent activity per capita correlates positively with higher CO₂ per capita at the cross-country level, suggesting that innovation proxied by patent applications reflects industrial development rather than decarbonization. European countries partially deviate from this trend, showing relatively high patent rates with moderate emissions, hinting at the effect of clean energy policies.

Nonparametric inference (Bootstrap CI for median CO₂ per capita):

A 10% random sample provides a reliable estimate of the population median CO₂ per capita. The 95% bootstrap confidence interval is narrow and encloses the true population median, demonstrating the validity of the bootstrap approach for skewed, non-normal data.